SMART HOME SECURITY SYSTEM BY USING ARDUINO UNO

Components Required

Arduino Uno
HC-SR04 Ultrasonic Sensor
PIR Motion Sensor
Red LED
Green LED
Buzzer
220 Ω Resistor
Jumper Wires
USB Cable
Breadboard

Working

The Smart Home Security System uses an Arduino Uno with a PIR motion sensor and an HC-SR04 ultrasonic sensor. The PIR sensor detects human motion, while the ultrasonic sensor measures the distance of nearby objects. The Arduino continuously reads both sensor values and activates the security system only when motion is detected and the object is within 30 cm. In this condition, the red LED turns ON, the green LED turns OFF, and the buzzer sounds an alarm. The system also displays the distance, motion status, and intrusion count on the Serial Monitor. This project can be used in homes, offices, classrooms, and laboratories for basic security and intrusion detection.

Connections

Component	**Arduino Uno Pin**
HC-SR04 VCC	5V
HC-SR04 GND	GND
HC-SR04 TRIG	D9

HC-SR04 ECHO	D10
PIR VCC	5V
PIR OUT	D2
PIR GND	GND
Red LED Anode (A)	D7 (through 220 Ω Resistor)
Red LED Cathode (C)	GND
Green LED Anode (A)	D6 (through 220 Ω Resistor)
Green LED Cathode (C)	GND
Pin 1:1	D8
Pin 1:2	GND

PROJECT LINK

https://wokwi.com/projects/468231892080308225